

A scalar counterexample to the signed polynomial mean-value trace conjecture

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Status. Candidate full counterexample; likely valid subject to human review. It disproves the polynomial half of the signed mean-value trace conjecture of Paulin, Mackey, and Tropp [1, 2].

1 The conjecture

For a Hermitian matrix X , write $X_+ = \max\{X, 0\}$ and $X_- = \max\{-X, 0\}$ by functional calculus. The polynomial assertion in Conjecture 7.3 of the source says that, for all Hermitian A, B, C , every positive integer q , and every $s > 0$,

$$\operatorname{tr}[C(A^q - B^q)] \leq \frac{q}{2} \operatorname{tr} \left[(s(A - B)_+^2 + s^{-1}C_+^2) |A|^{q-1} + (s(A - B)_-^2 + s^{-1}C_-^2) |B|^{q-1} \right]. \quad (1)$$

The technical report omits the trace symbol on the second right side, a typographical issue corrected in the published restatement [2]. The distinction disappears for the scalar counterexample below.

Conjecture 7.3 (Signed Mean Value Trace Inequalities). *For all matrices $A, B, C \in \mathbb{H}^d$, all positive integers q , and any $s > 0$ it holds that*

$$\operatorname{tr} [C(e^A - e^B)] \leq \frac{1}{2} \operatorname{tr} [(s(A - B)_+^2 + s^{-1}C_+^2)e^A + (s(A - B)_-^2 + s^{-1}C_-^2)e^B].$$

and

$$\operatorname{tr} [C(A^q - B^q)] \leq \frac{q}{2} [(s(A - B)_+^2 + s^{-1}C_+^2) |A|^{q-1} + (s(A - B)_-^2 + s^{-1}C_-^2) |B|^{q-1}].$$

This statement involves the standard matrix functions that lift the scalar functions $x_+ := \max\{x, 0\}$ and $x_- := \max\{-x, 0\}$. Extensive simulations with random matrices suggest that Conjecture 7.3 holds, but we did not find a proof.

2 An invertible odd-power counterexample

Theorem 1. *Inequality (1) is false already in dimension one, with the odd exponent $q = 3$. In particular, take*

$$A = \frac{1}{100}, \quad B = -1, \quad C = 1, \quad s = 1.$$

All three scalar matrices are invertible. The conjectured left side equals 1000001/1000000, while the right side equals 60603/200000000.

Proof. Since $A - B = 101/100 > 0$ and $C = 1 > 0$,

$$(A - B)_+ = \frac{101}{100}, \quad (A - B)_- = 0, \quad C_+ = 1, \quad C_- = 0.$$

The left side of (1) is

$$C(A^3 - B^3) = \frac{1}{100^3} - (-1)^3 = \frac{1000001}{1000000}.$$

Only the $|A|^2$ term survives on the right, so that side is

$$\frac{3}{2} \left[\left(\frac{101^2}{100^2} + 1 \right) \frac{1}{100^2} \right] = \frac{60603}{200000000}.$$

Hence the asserted inequality would require $1.000001 \leq 0.000303015$, which is false. $\square$

The failure is strict by more than three orders of magnitude. Continuity of both sides shows that it persists on an open neighborhood of this data; it is not caused by a zero eigenvalue or an endpoint convention. The choice $q = 3$ also lies in the odd-power family $q = 2p - 1$ used in matrix moment arguments.

3 Failure for every power above one

The same sign-allocation problem yields a uniform family.

Proposition 1. *For every integer $q \geq 2$, inequality (1) has a scalar counterexample. Take*

$$A = 0, \quad B = -1, \quad C = (-1)^{q+1}, \quad s = q.$$

The left side is 1. The right side is 0 when q is odd and $1/2$ when q is even.

Proof. Here $A - B = 1$, so $(A - B)_+ = 1$ and $(A - B)_- = 0$, while

$$C(A^q - B^q) = (-1)^{q+1}(-(-1)^q) = 1.$$

Because $q - 1 \geq 1$, the entire first summand on the right of (1) vanishes after multiplication by $|A|^{q-1} = 0$. If q is odd, then $C = 1$ and the second summand also vanishes. If q is even, then $C = -1$, so the second summand is $s^{-1}C_-^2|B|^{q-1} = 1/q$; multiplication by $q/2$ gives $1/2$. Both cases contradict the conjectured bound. $\square$

4 Scope and verification

Conjecture 7.3 asserts both an exponential inequality and the polynomial inequality (1). Theorem 1 gives a complete negative answer to the polynomial assertion and therefore disproves the conjunction as stated. It makes no claim about the separate exponential inequality. Nor does it conflict with the useful *unsigned* polynomial mean-value trace inequality proved elsewhere in the source, whose symmetric right side avoids this sign-allocation obstruction.

The included exact-arithmetic script verifies the displayed fractions and the family in Proposition 1. A bounded search on 13 August 2026 used the exact conjecture name and formula. It found the source, the 2016 published restatement, and later unsigned variants, but no earlier proof or disproof of the signed polynomial assertion. The novelty verdict is therefore *apparently new within the bounded search*, not exhaustive.

References

- [1] D. Paulin, L. Mackey, and J. A. Tropp, *Deriving matrix concentration inequalities from kernel couplings*, arXiv:1305.0612, Conjecture 7.3, p. 17 (2013).
- [2] D. Paulin, L. Mackey, and J. A. Tropp, *Efron–Stein inequalities for random matrices*, Ann. Probab. **44** (2016), no. 5, 3431–3473, Conjecture 13.4; arXiv:1408.3470.